



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES PAPER 1	DATE	24 NOVEMBER 2023
GRADE	11	MARKS	200
EXAMINER NAME	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
CLASS		DURATION	3 HOURS

INSTRUCTIONS:

1. This paper consists of 29 pages- 27 pages of questions and 2 pages of additional writing space. Please check that your question paper is complete.
 2. Answer all questions on the question paper.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. Any calculations need to be rounded off to TWO decimal places unless stated otherwise.
 7. It is in your own interest to write legibly and to present your work neatly.
-

QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	80	
2	40	
3	40	
4	40	
TOTAL: 200		

Question 1:

- 1.1. Select the name/ or term in Column II which best matches the description in Column I. Write the letter of the matching name/term in the appropriate space provided between the brackets. Each letter may only be used once. [10]

Column I**Column II**

- | | | |
|-------|--|----------------------------|
| [] | An indication of the variety of living things found in an environment. | A succession |
| [] | Process of gradual change of an ecological community over time. | B carrying capacity |
| [] | Linked food chains showing the energy flow in an ecosystem | C food web |
| [] | A group of organisms occupying the same area that can breed freely with one another. | D biotic factors |
| [] | Organisms that hunt and kill their food. | E abiotic factors |
| [] | Dividing vegetation into vertical layers. | F biodiversity |
| [] | Non-living components in the environment. | G community |
| [] | Area in which living and non-living things interact with each other. | H stratification |
| [] | Living components in the environment. | I ecosystem |
| [] | The number of individuals an area can sustain. | J population |
| | | K predator |

LIFE SCIENCES

1.2. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question in the space provided in the table below:

1.2.1.	1.2.2.	1.2.3.	1.2.4.	1.2.5.	1.2.6.	1.2.7.	1.2.8.	1.2.9.

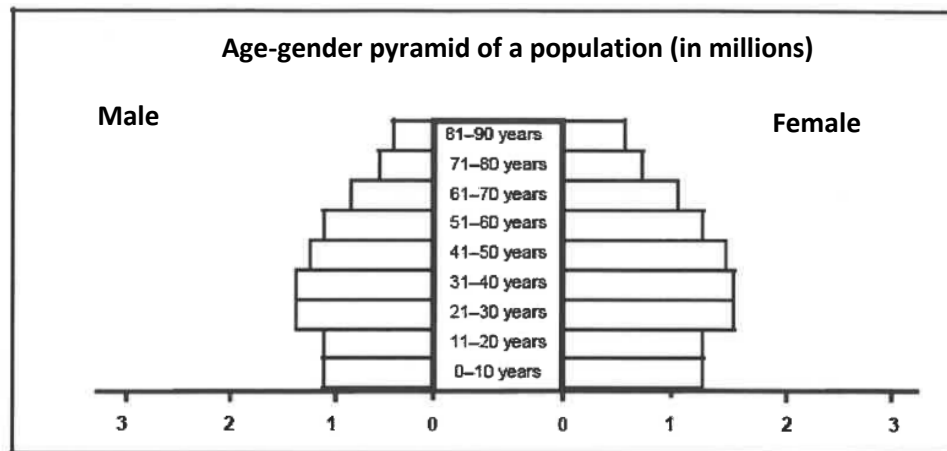
1.2.1. The pulmonary artery carries blood away from the (1)

- A. Right ventricle
- B. Right atrium
- C. Left ventricle
- D. Left atrium

1.2.2. Blood Type A will NOT agglutinate when mixed with (1)

- A. Type B
- B. Type A
- C. Type O
- D. None of the above

QUESTIONS 1.2.3. AND 1.2.4. ARE BASED ON THE PYRAMID BELOW:



1.2.3. The age-gender pyramid above is for a developed country since... (1)

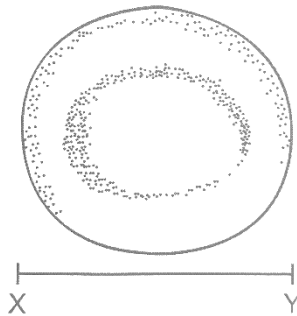
- A. The number of newborns is high.
- B. There are more young people than old people.
- C. There are more females than males in each age group.
- D. The life expectancy of the population is high.

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- 1.2.4. Which ONE of the following can correctly be deduced from the pyramid shown above? (1)
- A. There are less than 2 million people between 0 and 10 years.
 - B. There are more males than females in the age group 11 to 20 years.
 - C. The birth and death rates are about the same.
 - D. There are more females than males who are 50 years or older.
- 1.2.5. Living in a herd is of benefit to zebra as it: (1)
- A. Allows individuals to source more food.
 - B. Reduces competition as some zebra eat grass whilst other zebra eat leaves.
 - C. Allows the herd to avoid predators more easily.
 - D. Reduces competition for mates.
- 1.2.6. Which of the following combinations of characteristics, (i) to (iv), are true of a social species such as termites? (1)
- (i) Individuals live in groups
 - (ii) Individuals care for young that are not their own
 - (iii) All individuals of the colony have the same body structure
 - (iv) Not all individuals get to reproduce
- A. (i), (ii), (iii), (iv)
 - B. (i), (iii)
 - C. (i), (ii), (iv)
 - D. (ii), (iii)
- 1.2.7. Certain plants attract pollinators by generating heat at different times of the day. Different plant species heat up at different time of the day. This concept can be best described as: (1)
- A. A density independent factor
 - B. Resource partitioning
 - C. Asexual reproduction
 - D. Predation
- 1.2.8. Which of the following would NOT be classified as a density independent factor in decreasing population numbers? (1)
- A. Pollution from a surrounding factory
 - B. Oil spill in the ocean
 - C. Food availability
 - D. Hurricane

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- 1.2.9. A diagram of a human red blood cell is shown. The length of line XY on the diagram is 40mm. The actual width of the cell is 0.008mm.



What is the magnification of the diagram?

(2)

- A. x500
- B. x5 000
- C. x50 000
- D. x 500 000

[10]

- 1.3. Indicate whether each of the descriptions in COLUMN I applies to A ONLY, B ONLY, BOTH A AND B OR NEITHER of the items in COLUMN II. Write **A only**, **B only**, **both** or **neither** in the answer column. [5]

COLUMN I	COLUMN II	ANSWER
1.3.1. Heart muscle supplied with oxygen and nutrients	A. Pulmonary artery B. Coronary artery	
1.3.2. Capillaries	A. Microscopic B. Easy diffusion of metabolites in and out of cells	
1.3.3. Oxygenated blood	A. All arteries B. Venules	
1.3.4. Systolic blood pressure	A. Atria are contracted B. Ventricles are relaxed	
1.3.5. Smooth muscle layer of blood vessels	A. Vasoconstriction B. Vasodilation	

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- 1.4. An investigation was performed to determine the effect of exercise on heart rate. A learner (A) recorded her resting heart rate three times and recorded it. She then recorded her heart rate after one minute of walking on the spot, one minute jogging on the spot and one minute of running as fast as possible. She then did the same for her friend (B). Their results are recorded below.

Activity	Heart rate (bpm) Learner A	Heart rate (bpm) Learner B
Rest	54	68
Rest	52	66
Rest	52	67
Walking	72	80
Jogging	94	112
Running	132	150

Immediately after the last activity they measured their pulse rate every minute for 5 minutes. They recorded this below.

Time	Heart rate (bpm) Learner A	Heart rate (bpm) Learner B
1	110	137
2	95	120
3	72	102
4	54	84
5	54	72

- 1.4.1. Why do the heart rates of both learners increase after jogging on the spot for one minute? (2)

- 1.4.2. Which learner would you classify as the fittest? (1)

- 1.4.3. Give a reason for your answer to QUESTION 1.4.2. (1)

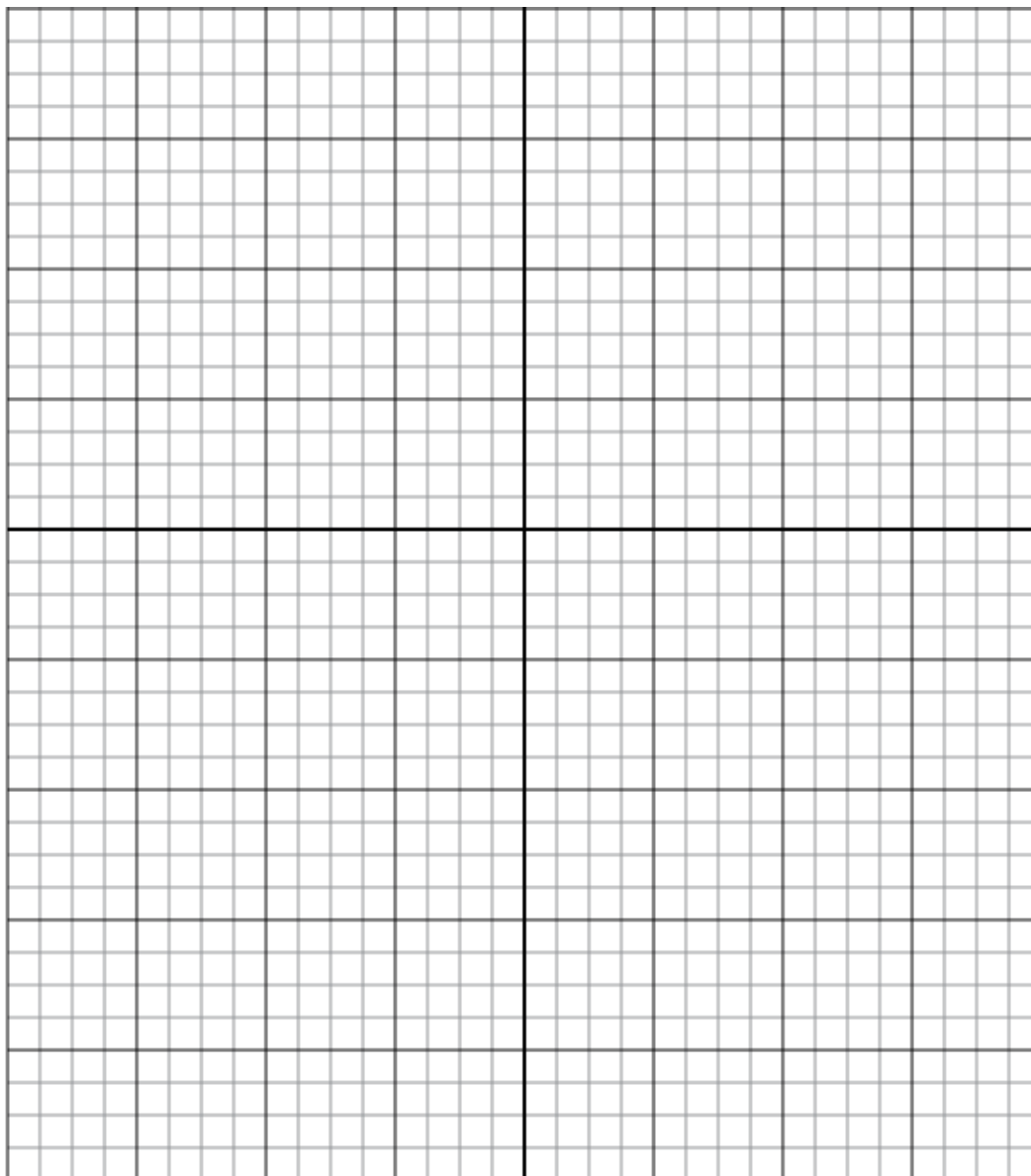
- 1.4.4. How long did it take for Learner A's heart rate to return to normal after running? (1)

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1.4.5. Name the dependent variable of this investigation. (1)

1.4.6. Name TWO controlled variables in this investigation. (2)

1.4.7. On the grid below draw a double line graph illustrating the results of **both** learners' heart rate for the five minutes after their last activity. (7)



[15]

1.5. The table below gives information about the blood flow in two people.

Person	Blood flow through the coronary arteries in cm ³ / minute
A	250
B	155

1.5.1. Which person do you think suffers from coronary artery disease? (1)

1.5.2. Give a reason for your answer to QUESTION 1.5.1. (2)

1.5.3. What substance causes blockages inside arteries such as coronary arteries? (1)

1.5.4. What could a person do to decrease the substance mentioned in QUESTION 1.5.3. in their diet? (1)

1.5.5. Calculate the difference in blood flow between person A and person B. (2)

1.5.6. Calculate the volume of blood flowing through the coronary arteries of person A in 1 hour. (2)

1.5.7. Coronary artery disease can be treated by:

- Inserting a stent
- Using a Coronary Artery Bypass Graft (CABG)

The table on the next page gives information about each method:

	Stent	CABG
Procedure	The patient is awake during the procedure. A small cut is made in the skin. A wire mesh is inserted into the coronary artery via a blood vessel in the arm or leg.	The patient is not awake during the procedure. The chest is cut open. A section of blood vessel from the arm or leg is removed. It is used to create a new channel for blood to bypass the blockage in the coronary artery.
When procedure is recommended	When only one blockage is present	When multiple blockages are present
Time spent in hospital after procedure	2-3 hours	At least 7 days
Recovery time after procedure	7 days	12 weeks
Risk of heart attack during procedure	1%	2%
Chance of failure within one year	40%	5%

a. Give TWO advantages of using a stent instead of CABG. (2)

- 1- _____

- 2- _____

b. Give TWO advantages of using CABG instead of a stent. (2)

- 1- _____

- 2- _____

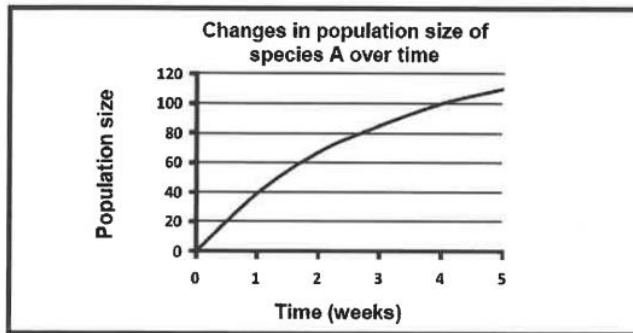
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LIFE SCIENCES

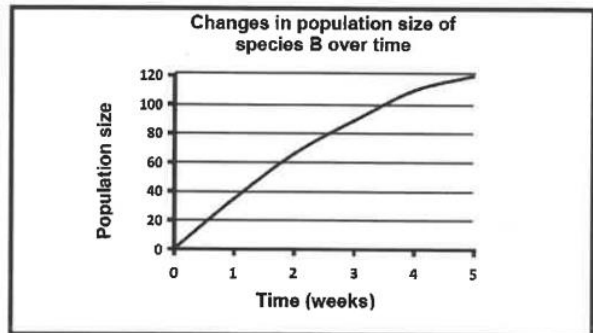
- 1.6. The growth patterns of two closely related species (**A** and **B**) that rely on the same food source were investigated. At first the two species were separated and then the two species were kept in the same habitat for the same period of time. In all cases the organisms were provided with a limited food supply.

The results were shown in three graphs (**A**, **B** and **C**) below.

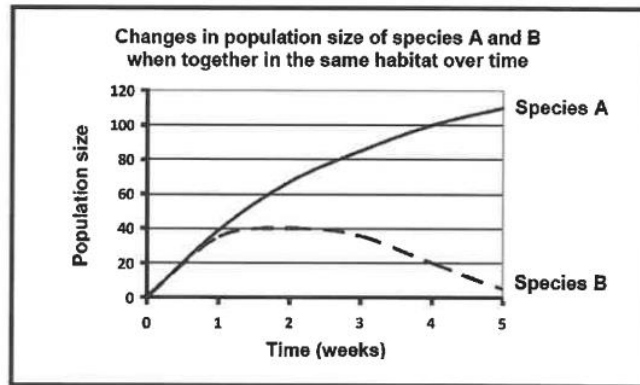
GRAPH A



GRAPH B



GRAPH C



- 1.6.1. Define the term 'population'. (2)

- 1.6.2. Name the independent variable in the above investigation. (1)

- 1.6.3. What is the difference in population numbers of species A and B after being in the same habitat for 4 weeks? (2)

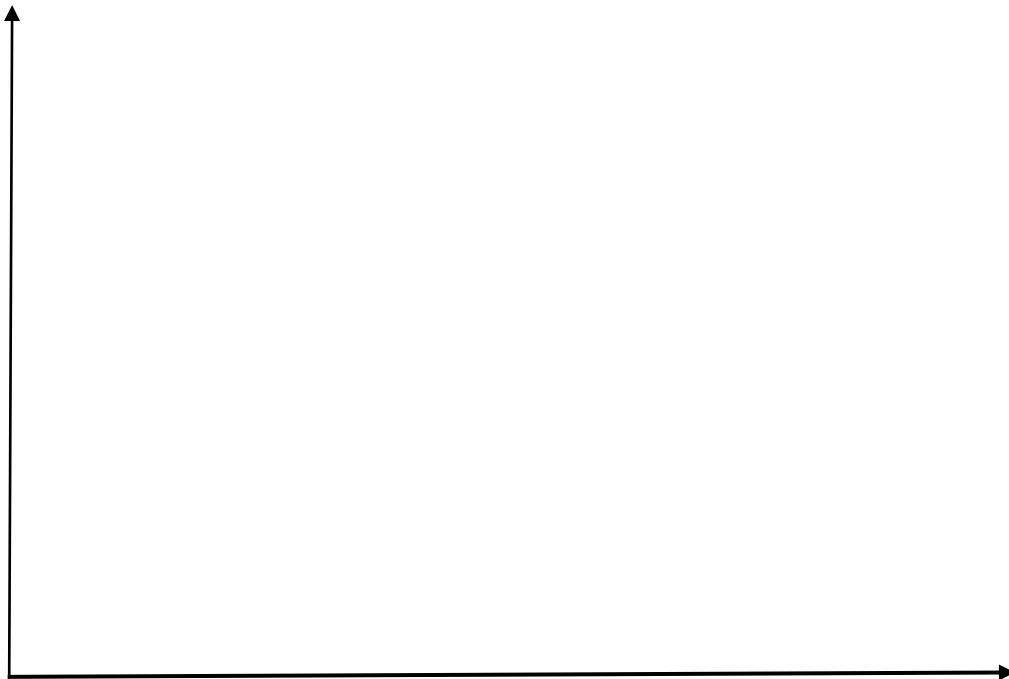
- 1.6.4. What principle is shown in graph C? (1)

- 1.6.5. Describe the growth patterns of species A and B when separated compared to when they are together in the same habitat. (4)

- 1.6.6. After some time, a natural predator of species A was introduced into the habitat.

- a. Differentiate between a predator and a prey. (2)

- b. On the set of axis below sketch the relationship between species A and it's predator over an extended period. (5)



[17]

- 1.7. Draw a line from the bones in the middle to the correct type of bone (left) and the location of it in the body (right). [8]

Type of bone		Location in the body
LONG BONE		WRIST
IRREGULAR BONE		LEG
FLAT BONE		SPINAL COLUMN
SHORT BONE		RIBCAGE

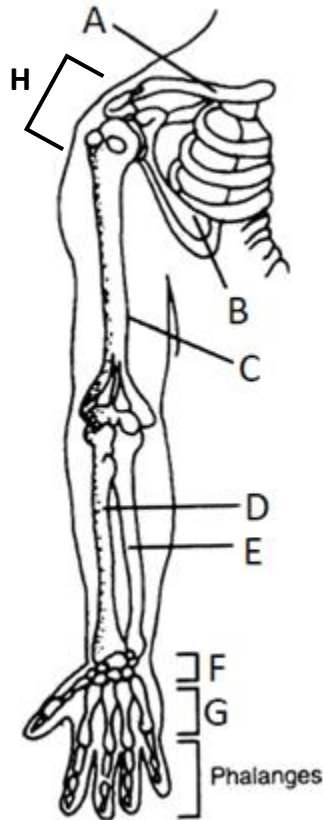
- 1.8. Name TWO types of skeletons other than that of a human. [2]

- 1- _____
2- _____

Total Question 1: 80 marks

Question 2

2.1. Study the diagram of a region of the human skeleton. Answer the questions that follow.



2.1.1. List TWO functions of the skeleton. (2)

1- _____

2- _____

2.1.2. What girdle is shown above? (1)

2.1.3. Give the LETTERS of the bones that make up the girdle mentioned in 2.1.2. (2)

2.1.4. Synovial joints are freely moveable joints that assist with movement.

Draw and label a diagram of a typical synovial joint in the space below.

(5)

2.1.5. Give the LETTER of a synovial joint in the diagram on the previous page.

(1)

2.1.6. Name the type of synovial joint AND the movement it allows mentioned in 2.1.5.

(2)

2.1.7. Give the names of bones B, D, E and G.

(4)

B- _____

D- _____

E- _____

G- _____

2.1.8. C is an example of a long bone. It contains both red and yellow bone marrow. What is the function of red marrow?

(2)

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2.2. Read the information below and answer the questions that follow.

The word arthritis means inflammation of the joints, and there are more than 100 forms of arthritis. One of the more common types is osteoarthritis.

Osteoarthritis occurs throughout the world and affects people differently, with joint damage developing over years or in some cases more quickly.

Osteoarthritis is a degenerative form of arthritis that often affects middle aged people, but can occur in young people who suffer from joint injuries.

Rheumatologists are doctors who are experts in diagnosing arthritis and other diseases of the joints, muscles and bones.

Warning signs of this disease are typically pain after an activity, limited range of motion and stiffness, and swelling and tenderness of one or more joints.

2.2.1. What is arthritis? (1)

2.2.2. What is a joint? (2)

2.2.3. Name the connective tissue that joins the bones at a joint. (1)

2.2.4. Give TWO characteristics of the tissue in 2.2.3. (2)

1-

2-

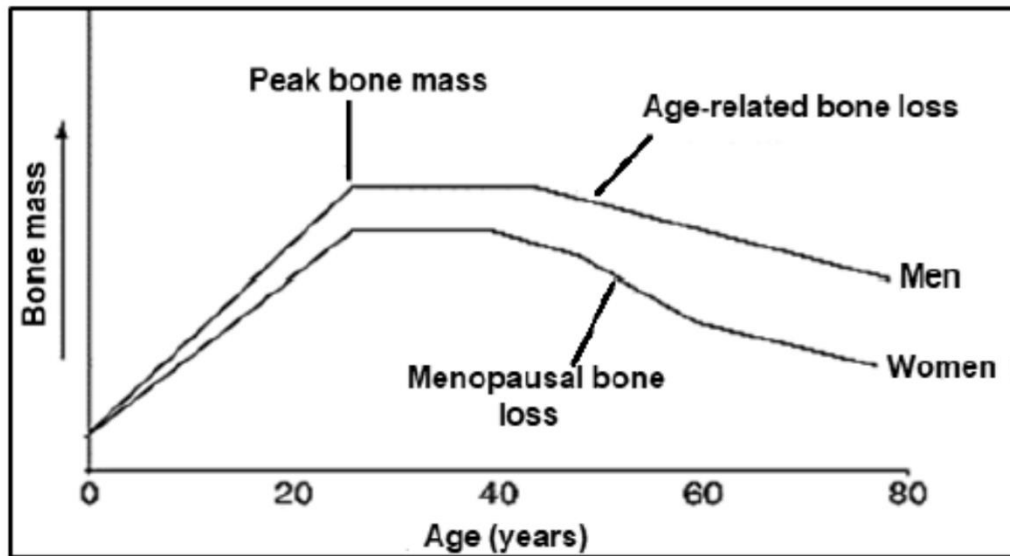
2.2.5. Describe the condition of the joint if a person has osteoarthritis. (3)

2.2.6. Suggest ONE reason why osteoarthritis is more common in middle-aged people. (2)

[11]

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2.3. Study the graph below and answer the questions that follow.



2.3.1. Name the disease in which bones bend due to poor bone formation in children. (1)

2.3.2. What vitamin is deficient in the above disease? (1)

2.3.3. When is the peak bone density formed in men and women? (2)

2.3.4. Give a heading for the graph above. (2)

2.3.5. Provide TWO conclusions from the graph above. (4)

- 1- _____

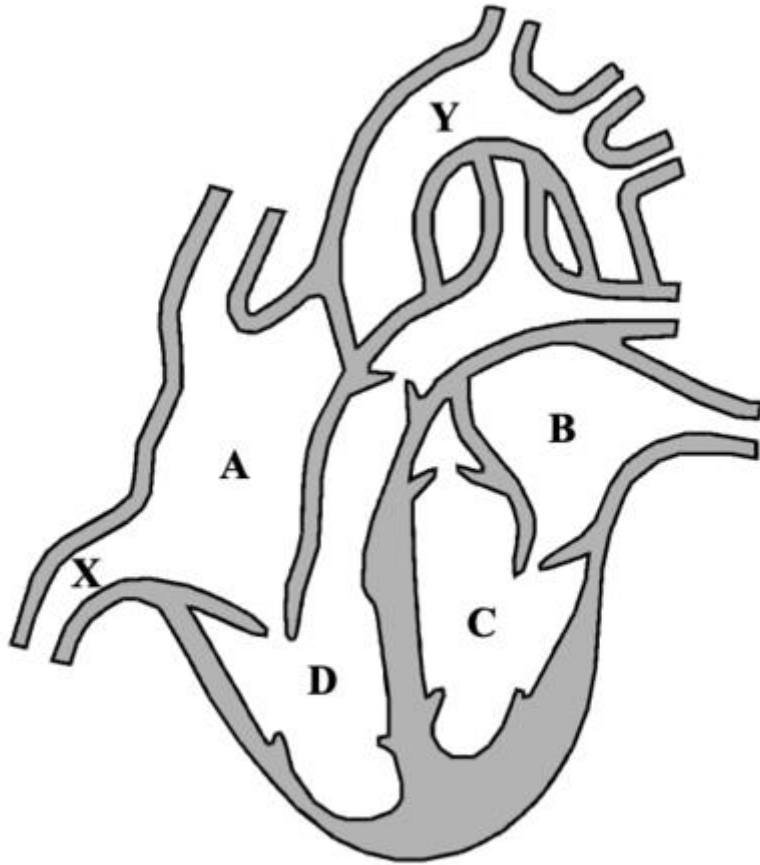
- 2- _____

[10]

Total Question 2: 40 marks

Question 3

3.1. The diagram below shows a section through the human heart.



3.1.1. Provide labels for A and C. (2)

A- _____

C- _____

3.1.2. Write down whether the following valves are open or closed when the pressure in chamber C is at its highest: (2)

a. Valve between chamber C and B. _____

b. Valve between chamber C and vessel Y. _____

3.1.3. Explain the advantage of having:

a. A large amount of elastic tissue in blood vessel Y. (2)

b. Valves in blood vessels such as blood vessel X.

(1)

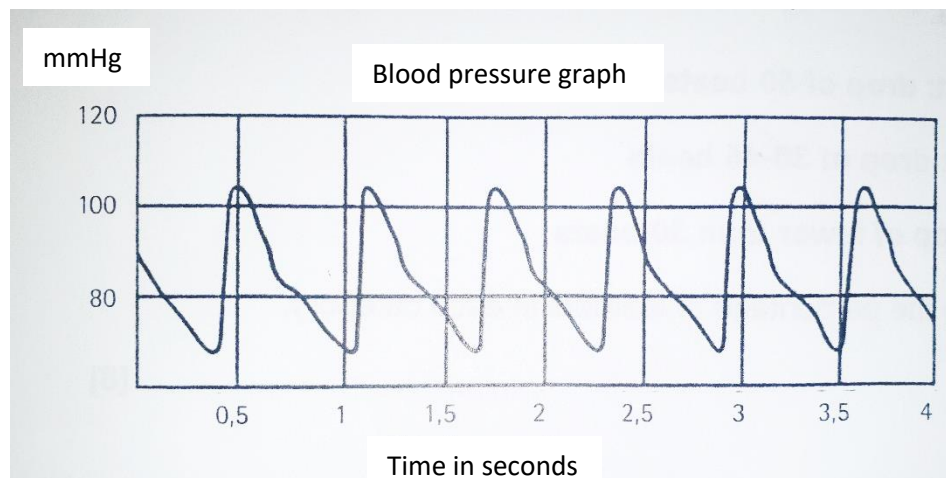
3.1.4. **On the diagram** on the previous page label the pulmonary artery and a pulmonary vein. (2)

3.1.5. Tabulate TWO structural differences between arteries and veins. (5)

3.1.6. A pacemaker is a small, battery-powered device that prevents the heart from beating too slowly. You need surgery to get a pacemaker. The device is placed under the skin near the collarbone.

On the diagram on the previous page label the heart's natural pacemaker. (2)

3.1.7. The blood pressure in vessel Y was recorded using an electronic pressure gauge. Here is the recording:



a. Why was vessel Y used to measure blood pressure? (1)

b. Explain why the pressure moves up and down. (2)

c. What factors might make the frequency of the heartbeat waves increase? (1)

d. What is the blood pressure of the person whose heartbeat is recorded here? (2)

[22]

3.2. The table below shows the recommended minimum heart rates that cyclists of different ages should maintain in order to either metabolise fat or improve their fitness.

Age	Minimum heart rate for metabolising fat (beats/min)	Minimum heart rate for improving fitness (beats/min)
10	136	168
20	130	160
30	123	152
40	116	Y
50	X	136
60	104	128

3.2.1. Predict the values of X and Y in the table above. (2)

X- _____

Y- _____

3.2.2. What is the independent variable in this investigation? (1)

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3.2.3. Explain what 'minimum heart rate for metabolising fat' means. (2)

3.2.4. Mention one fixed variable in this investigation. (1)

3.2.5. What conclusion can be made from the results above? (2)

[8]

3.3. *Blood doping is the practice of boosting the number of red blood cells in the bloodstream in order to enhance athletic performance. Because such blood cells carry from the lungs to the muscles, a higher concentration in the blood can improve an athlete's aerobic capacity (VO_2 max) and endurance.*

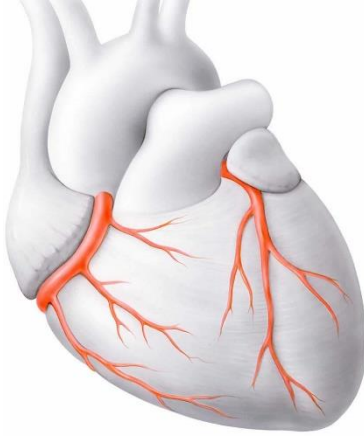
3.3.1. Explain one structural adaptation of red blood cells that allows them to maximise their oxygen carrying capacity. (2)

3.3.2. It is calculated that blood doping on average can increase performance by 12% based on a correlating number of red blood cells. If blood normally contains 250 000 000 cells, calculate how many red blood cells would be found on average in doped blood. Show all working. (3)

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It is well known that blood doping, by thickening the blood, leads to an increased risk of several deadly diseases such as heart disease (also known as coronary artery disease (CAD)) and stroke.

- 3.3.3. **On the diagram below** mark with an **X** where a blockage would occur that could cause a major, and possibly fatal, heart attack. (1)



- 3.3.4. Name TWO risk factors of CAD that can be controlled. (2)

- 3.3.5. Name ONE risk factor of CAD that cannot be controlled. (1)

- 3.3.6. Identify ONE symptom of a stroke. (1)

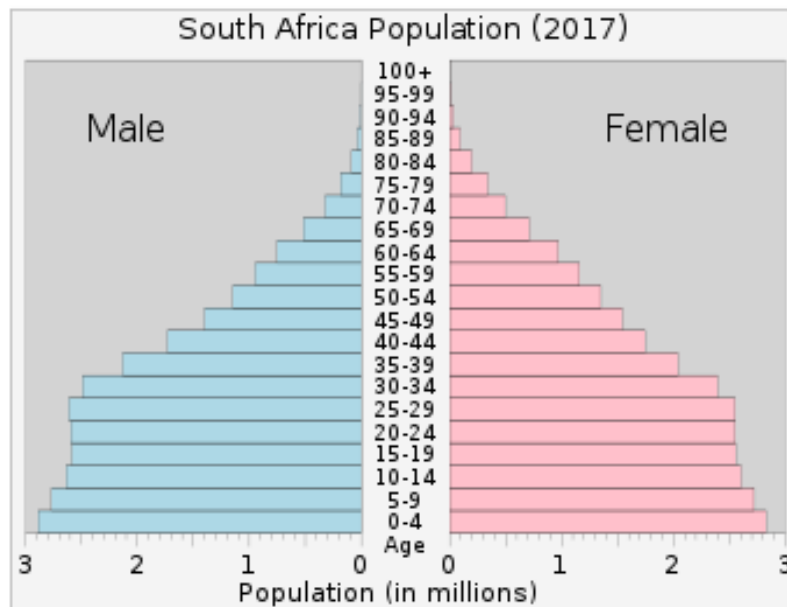
Total Question 3: 40 Marks

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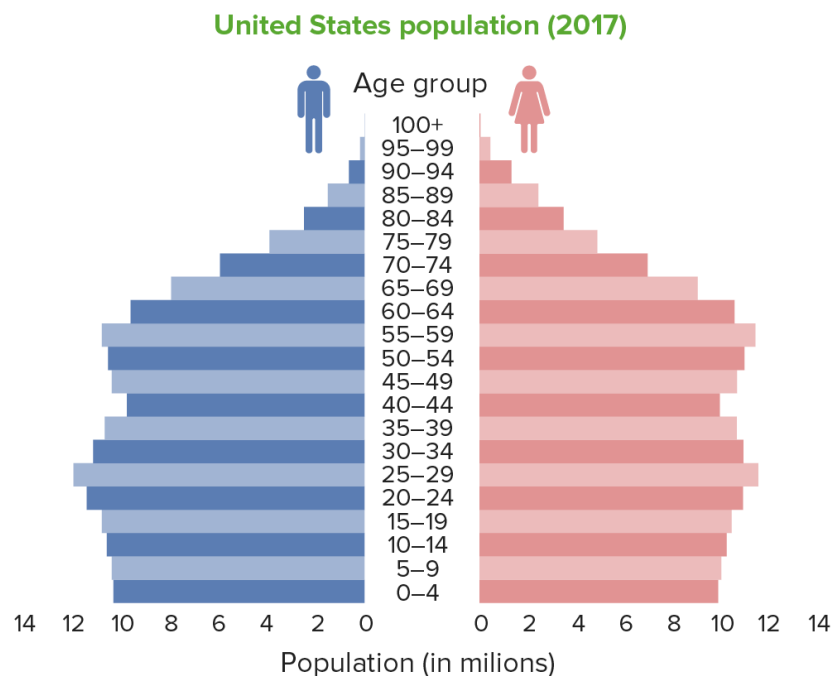
Question 4

- 4.1. Study the Age-gender pyramids representing two different countries and answer the questions that follow.

A



B



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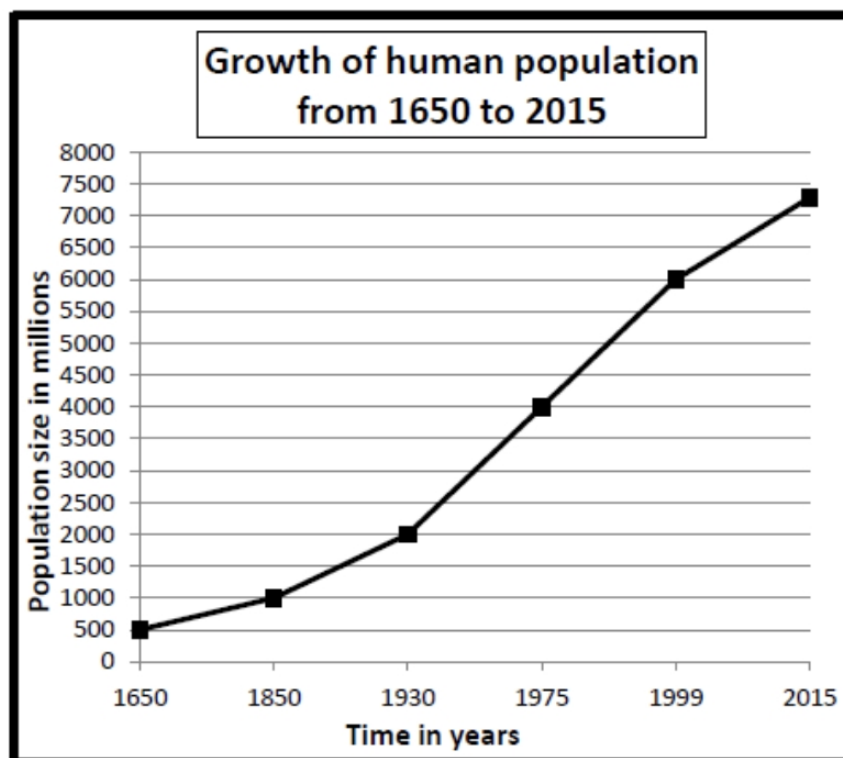
4.1.1. Name the source of information required to plot the above graph. (1)

4.1.2. Which population pyramid (A or B) shows an increasing population? (1)

4.1.3. Explain your answer to QUESTION 4.1.2. (4)

4.1.4. Describe the trend of population growth in Graph B. (3)

4.1.5. The graph below shows the growth of the human population from 1650 to 2015.



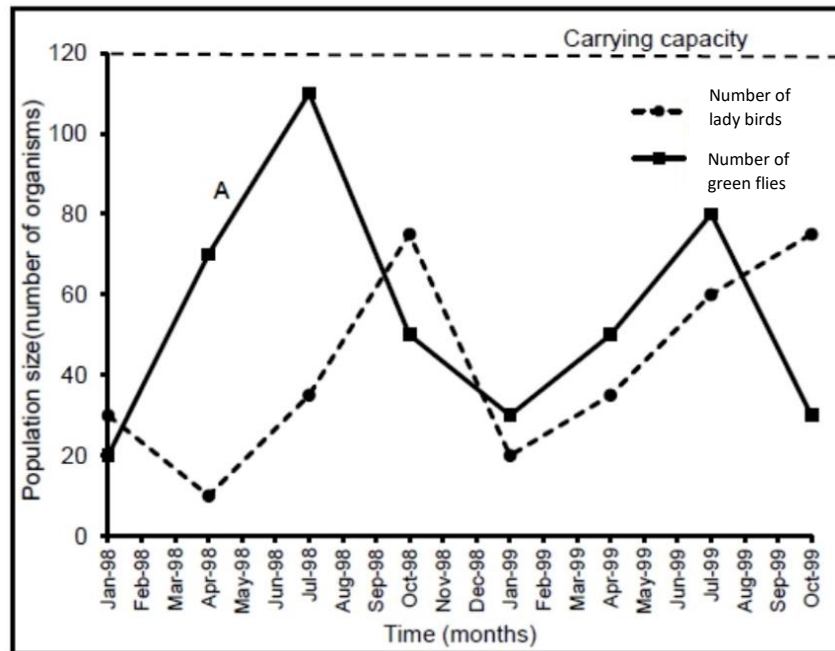
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- a. What growth form is shown by humans from 1930 to 1999? (1)
- _____
- b. Give any TWO reasons for the slow population growth from 1999 to 2015. (2)
- 1- _____
- _____
- 2- _____
- _____
- c. **Current global population of over 7 billion is already two to three times higher than the sustainable level. Several recent studies show that Earth's resources are enough to sustain only about 2 billion people at a European standard of living.**
- i. Define the term 'carrying capacity'. (2)
- _____
- _____
- ii. How have humans managed to continually increase the earth's carrying capacity? (2)
- _____
- _____
- iii. Predict the fate of humans if the current population growth is sustained without check. (1)
- _____
- iv. Suggest a practical solution to reduce the accelerated growth of the human population. (1)
- _____
- _____

[18]

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4.2. The graph below represents an interaction between two organisms in an ecosystem.



4.2.1. What type of relationship is shown in the graph? (1)

4.2.2. Identify the prey population in the above graph. (1)

4.2.3. Provide a reason for your answer to QUESTION 4.2.2. (1)

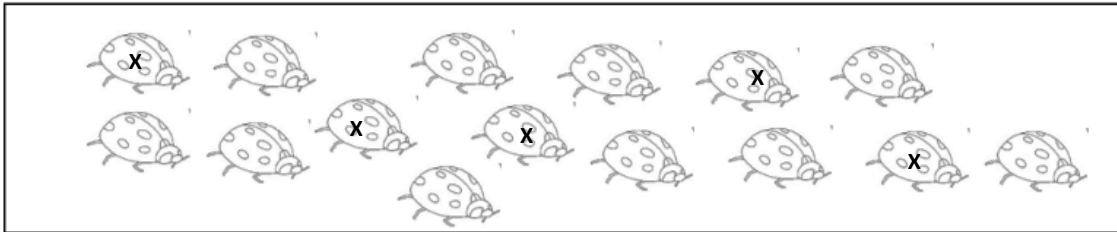
4.2.4. Explain why the chances of lady birds and green flies reaching the carrying capacity is limited. (3)

4.2.5. Explain the changes in green fly population from July 1998 to October 1998. (3)

[9]

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- 4.3. Ashleigh catches lady bugs at random and marked each one with an X and returned them to her garden. After one week, a second sample was collected. The diagram below shows the lady bugs that were caught during the second selection.



- 4.3.1. Using a formula, it is determined that there are 60 lady bugs in the population. How many lady bugs were there in the first sample? (3)

- 4.3.2. Name TWO precautions Ashleigh needs to take when completing this investigation. (2)

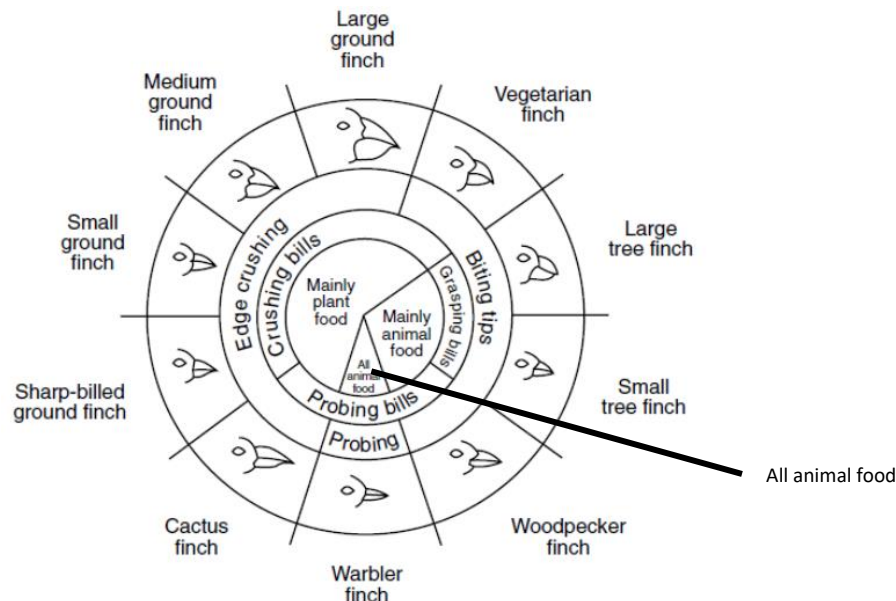
- 4.3.3. How could Ashleigh improve the accuracy and reliability of her investigation. (1)

- 4.3.4. Why is the mark-recapture method the best method to use to determine the lady bug population? (2)

[8]

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- 4.4. Fourteen finch species, with beaks of different shapes and sizes, originated from one common ancestor by means of evolution. The different shapes and sizes of their beaks determine the type of food they eat. Study the diagram below and answer the questions that follow.



- 4.4.1. What structural feature determines the different species of finch? (1)

- 4.4.2. In each of the following, CIRCLE only the letter that corresponds: (2)

- a. Which two finches will compete the least with each other for food:

- A small ground finch and large ground finch
- B large ground finch and sharp-billed ground finch
- C small tree finch and medium ground finch
- D vegetarian finch and small ground finch

- b. The only exclusively carnivorous finch has a bill that is adapted as follows:

- A only probing
- B probing and edge crushing
- C probing and biting tips
- D biting tips and edge crushing

- 4.4.3. Would the large tree finch and large ground finch be able to co-exist on the same island? Give a reason for your answer. (2)

[5]

Total Question 4: 40 marks

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.